Multiply and Divide by 10, 100 and 1000

Australian Curriculum v9.0 · Year 5
Maths
(Year 4 review skill)
AC9M4N05

Back to School 2027 Warm-Up

Name: _____ Date: _____

LEARNING INTENTION

I can multiply and divide whole numbers by 10, 100 and 1000, and use these ideas to work with multiples such as 30 and 600, using what I know about place value.

I CAN...

- ✓ multiply by 10, 100 and 1000
- ✓ divide by 10, 100 and 1000
- ✓ use a fact like 3 × 2 to work out 30 × 20

KEY STRATEGY AND WORKED EXAMPLE

Do not just "add a zero". Think about **place value**: multiplying makes each digit worth 10 times more, so it moves one place left. Dividing makes each digit worth 10 times less, so it moves one place right.



34 ones × 10 = 34 tens = 340. Every digit moves 1 place to the LEFT.
Dividing by 10 does the opposite: every digit moves 1 place to the RIGHT.

Multiples of 10: 6 × 40 → 6 × 4 = 24, then × 10 = 240 | 4 500 ÷ 100 = 45

A. TRY TOGETHER – USE THE STRATEGY

1. 7 × 10 = _____
2. 45 × 10 = _____
3. 36 × 100 = _____
4. 800 ÷ 10 = _____
5. 5 × 30 = _____
6. Which way do the digits move when you divide by 100? _____

B. CORE PRACTICE – SHOW YOUR THINKING

7. 63 × 10 = _____
8. 8 × 100 = _____
9. 125 × 10 = _____
10. 47 × 100 = _____
11. 9 × 1000 = _____
12. 6 000 ÷ 10 = _____
13. 3 500 ÷ 100 = _____
14. 72 000 ÷ 1000 = _____
15. 4 × 60 = _____
16. 7 × 800 = _____
17. 900 ÷ 30 = _____
18. 2 400 ÷ 60 = _____

C. CHALLENGE – REASON AND EXPLAIN

19 The canteen orders **20 boxes** of apples. Each box holds **30 apples**. How many apples is that? Show how a basic fact (3 × 2) helps you.

20 **Error Analysis:** Mia says, "450 ÷ 10 = 4 500 because you add a zero." Explain her mistake. What is the correct answer, and why?

21 A bakery packs **6 000 cupcakes**. (a) How many boxes are needed if each box holds 100? (b) How many boxes if each box holds 10? What do you notice about your two answers?

Watch Out! Before you write an answer, ask yourself: am I making the number bigger or smaller? Which way should the digits move?

💡 **Parent Tip:** Draw a simple Hundreds / Tens / Ones chart on paper and slide counters or digit cards left and right. Ask: "What happened to each digit?" Talking about the movement matters more than the answer.

🎯 My Term 1 Goal: This term I want to get better at _____

Answer and Teacher Page

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ANSWERS

1. 70

2. 450

3. 3 600

4. 80

5. 150 (5 × 3 = 15, then × 10)

6. To the right (one place for each zero: 2 places for ÷ 100)

7. 630

8. 800

9. 1 250

10. 4 700

11. 9 000

12. 600

13. 35

14. 72

15. 240

16. 5 600

17. 30

18. 40

19. 600 apples. 3 × 2 = 6, then 30 × 20 is 100 times bigger (× 10 × 10), so 600.

20. Dividing makes the number smaller: the digits move right. 450 ÷ 10 = 45. Mia multiplied instead of dividing.

21. (a) 60 boxes (b) 600 boxes. Boxes of 10 means 10 times more boxes, because each box holds 10 times fewer cupcakes.

TEACHER NOTES

Common misconception: "adding a zero". Watch for students who write 450 ÷ 10 = 4 500 or 3.5 × 10 = 3.50. Return to the place value chart.
Support: use a Hundreds / Tens / Ones chart with digit cards for questions 1 to 6.
Challenge marking: Q19 full marks for a correct answer plus a reference to the 3 × 2 fact and the two zeros. Q20 and Q21 look for the words "digits move", "10 times smaller" or "10 times more".
Australian Curriculum v9.0: AC9M4N05 (Year 4 review): solve problems involving multiplying or dividing natural numbers by multiples and powers of 10 without a calculator, using the multiplicative relationship between the place value of digits.